

AI-Powered Customer Support & Ticket Platform

Project Report & Technical Documentation

1. Executive Summary

This project presents a modern, production-ready AI Customer Support and Ticket Management System developed using Python 3.12, Flask, Flask-SQLAlchemy, Flask-Login, and Bootstrap 5. It delivers zero-friction guest AI chat assistance alongside automated ticket escalation.

2. Student & Project Metadata

Student Name: Pamulapati Karthik

Roll Number: 22N81A05K2

Department: Computer Science & Engineering (CSE)

College: Sree Dattha Group of Institutions

Guide: Dr. A. Ramesh Kumar (HOD & Professor)

3. Key Objectives & Outcomes

- Instant guest AI assistance without customer login barriers.
- Automated pre-LLM Knowledge Base RAG lookup reducing operational costs.
- Auto-generating unique Ticket IDs (TICK-XXXXXXX) when AI confidence is low.
- Single-auth Admin Console for ticket management, Chart.js analytics, and CSV exports.